Computer

Class 6 (English Medium)

Solved Exercises — Sindh Textbook Board

6 lessons — 162 questions

Introduction to Computer

[1-1] Which of the items below is used for storing information?

(a) Microprocessor

✓ (b) Memory card

(c) Router

(d) Motherboard

[1-2] Computers of which of the eras below used the most electricity?

✓ (a) First era

(b) Second era

(c) Third era

(d) Fourth era

[1-3] Computers of which of the eras below are more powerful and useful?

(a) First era

(b) Second era

(c) Third era

✓ (d) Fourth era

[1-4] Which of the devices below can be used both for sending data in and for getting the result out?

(a) Plotter

(b) Monitor

✓ (c) Interactive board

(d) Track ball

[1-5] Which of the devices below can store the most data?

(a) CD

✓ (b) Blu-ray Disc

(c) Floppy Disk

(d) DVD

[2-1] Network switch is a communication device.

[2-2] _____ are the devices used for getting information from the computer.

Answer: Output devices

[2-3] _____ are the devices used for exchanging data between networks.

Answer: Routers

[2-4] _____ is the type of software used to manage and control the computer.

Answer: System software

[2-5] _____ is the type of software used to help the user in everyday tasks.

Answer: Application software

Question: [3-1] What is a computer?

Answer: A computer is an electronic data processing machine that accepts data (input), processes it, and gives information (output). For example, 2 + 4 is typed on the keyboard, the computer processes it, and shows 6.

Question: [3-2] What is the difference between computer hardware and software?

Answer: Hardware is the physical parts of the computer that can be touched, such as the keyboard, mouse and display unit. Software consists of programs (a set of instructions) that tell the computer what to do, such as MS Windows, Linux and Paint.

Question: [3-3] Explain the difference between input devices and output devices.

Answer: Input devices send data to the computer for processing, such as

the keyboard, mouse and scanner. Output devices deliver the processed information to the user, such as the monitor, printer and speaker.

Question: [3-4] Write the names of five storage devices.

Answer: Five storage devices are: (1) Hard Disk Drive (2) CD (3) DVD (4) Blu-ray Disc (5) USB flash drive.

Question: [3-5] Describe the processing devices.

Answer: Processing devices are the part of the computer that processes data, turns it into information, and helps store it in memory. They include the microprocessor (CPU), ROM, RAM and the motherboard.

Question: [3-6] Describe the key features of the different generations of computers.

Answer: First generation (1940–1950): vacuum tubes, very large, expensive, much heat and electricity. Second generation (1950–1960): transistors, smaller, faster, more reliable. Third generation (1960–1970): integrated circuits, small size, low cost, more speed. Fourth generation (1970 to now): microprocessor, personal computers, graphical interface, Windows, internet. Fifth generation (present and future): artificial intelligence, quantum computing, Internet of Things.

Question: [3-7] How can ICT be used in the field of education?

Answer: In education ICT is used like this: educational games help students learn new concepts; through online learning (such as Khan Academy and Virtual University) students can study different subjects; the interactive whiteboard helps the teacher teach with videos and pictures; and computers are also used for research, simulation and collaborative work.

Question: [3-8] How can ICT be used in the field of industry?

Answer: In industry ICT is used like this: robots help make and assemble things in factories; computer-controlled machines make work accurate and

fast; and the computer is used to manage the record and supply of goods in the warehouse.

Question: [3-9] What benefits can students get from ICT?

Answer: Students can get these benefits from ICT: wide access to educational resources, learning through interactive games and video-sharing apps, learning from platforms such as Khan Academy and Duolingo, communication through email and social media, cooperation in joint projects, and creative work with tools such as Paint and Canva.

Question: [3-10] What are the disadvantages of ICT for beginners?

Answer: The disadvantages of ICT for beginners are: distraction — wrong use or lack of supervision affects students' performance; health problems — eye strain, lack of sleep and physical health problems; cyber security risks — virus, malware, hacking and phishing attacks; and dependence — overuse of devices makes a person dependent on them.

Question: [4-1] Explain the function of a router.

Answer: A router connects two or more networks together. For example, a home router connects the home network to the internet service provider (ISP).

Question: [4-2] Explain the function of a LAN card.

Answer: A LAN card (Network Interface Card) connects the computer to a Local Area Network, through which the computer connects to other computers and communication devices.

Question: [4-3] Explain the function of a QR code scanner.

Answer: A QR code scanner reads the data of a barcode or QR code and sends it to the computer for processing.

Question: [4-4] Explain the function of a touch pad.

Answer: On a touch pad, the cursor on the screen is moved simply by the movement of a finger or stylus (pen). A drawing pad helps in drawing, and

a game pad helps in playing games.

Question: [4-5] Explain the function of a headphone.

Answer: A headphone is an audio output device that works like a speaker — it delivers the sound of the computer's information directly to the listener's ears.

Question: [4-6] Explain the function of a projector.

Answer: A projector connects the computer to a screen and shows text, graphics and video information on a large screen.

Question: [4-7] Explain the function of a microprocessor.

Answer: The microprocessor is the processing brain of the computer — it processes data. It equals thousands of integrated circuits on a single chip, gives fast performance, is small in size, and produces little heat.

Question: [4-8] Explain the function of a motherboard.

Answer: The motherboard is the main board of the computer, on which all the hardware parts of the computer are connected, such as the power supply, microprocessor, RAM, ROM and storage devices.

Question: [4-9] Explain the function of a flash drive.

Answer: A USB flash drive is used to store digital information. It is small and portable, so files can easily be transferred from one place to another.

Question: [5-A] Identify the device shown in the picture and write its name: a small USB device labelled Bluetooth with a small cable attached.

Answer: Bluetooth adapter — this is a communication device that enables the computer to exchange data wirelessly with nearby devices.

Question: [5-B] Identify the device shown in the picture and write its name: a card with an antenna attached, which fits inside the computer.

Answer: Network Interface Card (LAN card) — it connects the computer to a network.

Question: [5-C] Identify the device shown in the picture and write its

name: a large printing device printing a map or drawing on large paper.

Answer: Plotter — this is an output device that prints maps and large pictures on large paper, plastic or flex.

Question: [5-D] Identify the device shown in the picture and write its name: a green circuit board with many parts and slots fitted on it.

Answer: Motherboard — this is the main board of the computer, on which all the hardware parts are connected.

Question: [5-E] Identify the device shown in the picture and write its name: a square screen standing on a stand.

Answer: Monitor — this is the basic output device, which shows the computer's text, picture and video information.

Question: [5-F] Identify the device shown in the picture and write its name: a device printing a coloured picture on paper.

Answer: Printer — this is an output device that prints text and graphics information on paper.

Question: [5-G] Identify the device shown in the picture and write its name: a device fitted above an open book, capturing the image of the book's pages.

Answer: Scanner — this is an input device that converts the picture of paper or a book into digital data and sends it to the computer.

Question: [5-H] Identify the device shown in the picture and write its name: a hand-held controller with buttons and joysticks on it.

Answer: Game pad (game controller) — this is an input device used for playing games.

Question: [5-I] Identify the device shown in the picture and write its name: a black box with three antennas fitted on it.

Answer: Router — this is a communication device that connects two or more networks together.

Question: [7-A] What type of device is shown in the picture? A white printing device (HP).

Answer: Output device — a printer prints text and graphics information on paper, so it is an output device.

Question: [7-B] What type of device is shown in the picture? A small memory card (microSD, 8 GB).

Answer: Storage device — a memory card is used to store data and information.

Question: [7-C] What type of device is shown in the picture? Two small black memory chips (ROM).

Answer: Processing device — ROM chips hold the instructions for starting the computer and belong to the processing part.

Question: [7-D] What type of device is shown in the picture? A mouse with a red ball (track ball) on top.

Answer: Input device — moving the ball of the track ball mouse moves the cursor, so it is a device for sending data in.

Question: [7-E] What type of device is shown in the picture? A hard disk drive and an SSD.

Answer: Storage device — both the hard disk and the SSD are used to store data and programs permanently.

Question: [7-F] What type of device is shown in the picture? A hand-held barcode scanner.

Answer: Input device — a barcode scanner reads the barcode data and sends it to the computer.

Question: [7-G] What type of device is shown in the picture? A network switch (D-Link).

Answer: Communication device — a network switch is used to connect computers together in a network.

Question: [7-H] What type of device is shown in the picture? A microprocessor chip (Intel Pentium 4).

Answer: Processing device — the microprocessor is the processing brain of the computer, which processes data.

Question: [7-I] What type of device is shown in the picture? A printer pushing paper out (HP).

Answer: Output device — a printer prints information on paper, so it is an output device.

Question: [7-J] What type of device is shown in the picture? Two green memory strips (RAM).

Answer: Processing device — RAM is temporary memory, in which running programs and data stay during processing.

Question: [8-1] Group activity: Draw some parts of the computer hardware on a drawing sheet.

Answer: Draw these parts on the drawing sheet and write their names: keyboard (for typing data), mouse (for moving the cursor), monitor (for showing information), system unit (for processing) and printer (for printing on paper).

Question: [8-2] Group activity: Prepare a poster presentation on any ICT topic.

Answer: Choose a topic, for example "ICT in education" — write the title on the poster, write 3–4 short points (online learning, educational games, interactive board), add matching pictures, write a short conclusion at the end, then present it in class.

Digital Skills / Use of Computer

What is the main role of the bottom strip (taskbar) in Windows 10?

(a) Managing the machine's resources

(b) Providing an environment for the user for files and folders

✓ (c) Organising the different Windows features and giving quick access to them

(d) Protecting the machine from viruses

What is File Explorer used for in Windows 10?

(a) Managing the machine's resources

(b) Organising the different Windows features and giving quick access to them

✓ (c) Viewing and organising files and folders

(d) Protecting the machine from viruses

Where do deleted files stay stored for some time? What is that place called?

(a) Windows taskbar

✓ (b) Recycle Bin

(c) System tray

(d) Windows desktop

What is the benefit of the Quick Access Toolbar in MS Paint?

(a) Making a picture or drawing

(b) Inserting shapes such as circles and rectangles

(c) Making a picture or artwork with the freehand tool

✓ (d) Getting access to everyday features such as Save and Undo

What is the main purpose of a search engine on the internet?

✓ (a) Helping users find information about a particular topic

(b) Helping users create a particular web page

(c) Helping users download films

(d) Helping users play online games

In Windows 10, by right-clicking on a selected file or folder, one can make its shortcut by choosing _____ from the menu that appears.

Answer: Create Shortcut.

_____ is a simple writing program present in Windows, which is included in every edition of Windows.

Answer: Notepad.

The _____ palette holds many colours to choose from for filling colour in a drawing.

Answer: Colour.

In MS Paint, the _____ keyboard shortcut is pressed to take back the last action performed.

Answer: CTRL+Z.

A search engine helps the user find the required information or data on the basis of given _____ or sentences.

Answer: Keywords.

Question: What is meant by an operating system?

Answer: An operating system is a program that controls and manages all the parts and programs of the computer, so that using the computer becomes simple and easy for a person. Its main functions are: managing resources, providing an environment to the user, running programs, managing files, and security.

Question: What work does the taskbar perform in Windows 10?

Answer: The taskbar is a horizontal strip at the bottom of the desktop. It organises the different Windows features and gives quick access to them, such as: opening applications, searching, viewing the date and time,

changing settings, and so on.

Question: Write the method for making the outline of a drawing and filling it with colour in Microsoft Paint.

Answer: First choose a shape from the Tools box and draw the outline of the drawing with the mouse. Then choose the "Fill with color" tool from the Tools box, pick the required colour from the colour palette, and click inside the shape, so the inside gets filled with colour. To change the outline or the inside colour, select the Colour 1 or Colour 2 box and choose a new colour.

Question: What safety points should be kept in mind while using the internet? Explain.

Answer: Do not generally share your personal information such as your name, phone number, bank account, card details, address and password with anyone. Before clicking any web link, confirm that it is genuine, so that you can stay safe from a harmful file or fraud. Always behave online with respect and kindness towards others. If anything online feels strange or wrong, tell your parents or teacher at once.

Question: Describe the work of a search engine.

Answer: Search engines are websites that help find information on the internet quickly and easily. The user types a keyword or a simple sentence in the search bar, and the search engine shows a list of links to matching websites. Examples are Google, Bing, Yahoo, Ask and DuckDuckGo.

Question: Match the pictures in Column A with the names in Column B. Column A: 1) Paint brush icon, 2) Edge icon, 3) Windows logo, 4) The three window control buttons, 5) Recycle Bin icon. Column B: (a) Windows 10 Start menu / Start button, (b) Recycle Bin, (c) Window controls, (d) Web browser, (e) Brush.

Answer: 1) Paint brush icon — (e) Brush; 2) Edge icon — (d) Web browser; 3) Windows logo — (a) Windows 10 Start menu / Start button; 4) The three

window control buttons — (c) Window controls; 5) Recycle Bin icon — (b) Recycle Bin.

Question: Write the name of the icon below and explain it briefly: the "Type here to search" box.

Answer: Name: Search box. Explanation: This box is used to search for files, folders, applications, settings and things on the web; as soon as a word is typed in it, the matching results appear.

Question: Write the name of the icon below and explain it briefly: the icon showing a basket-like picture.

Answer: Name: Recycle Bin. Explanation: This is a special folder in Windows in which deleted files and folders stay safely stored for some time, so that they can be brought back if needed.

Question: Write the name of the icon below and explain it briefly: the right-hand part of the taskbar, where the time, date, battery, sound and network symbols appear.

Answer: Name: System tray or notification area. Explanation: This is the right-hand part of the taskbar, which shows system states such as the battery level, sound, network connection, date and time, and gives access to their settings.

Question: Write the name of the icon below and explain it briefly: the yellow folder icon.

Answer: Name: Folder. Explanation: A folder is a box or container for organising and storing files; different kinds of files can be kept in separate folders.

Question: Write the name of the icon below and explain it briefly: the colourful circle icon (Google Chrome).

Answer: Name: Google Chrome (web browser). Explanation: This is a web browser through which websites on the internet are opened and viewed; it

is one of the most widely used browsers in the world.

Algorithmic Thinking and Problem Solving

Question: 1. The teacher tells the class several problems; on what basis will you tell which problem is simple and which one is complex?

Answer: A simple problem is one solved just by joining steps in order, needing no expert, e.g. making a cake; a complex problem is one needing many stages, thinking and different methods, e.g. finding the LCM of three numbers or checking whether a number is prime.

Question: 2. Take one problem from your daily life and describe its solution through the six problem-solving stages.

Answer: Example — a birthday gift for a friend: 1. Identify the problem — which gift to give the friend; 2. Gather information — think about the friend's likes, the budget and the shops; 3. Review possible solutions — a book, a game, a gift voucher or a photo album; 4. Choose the best solution — a special photo album of the friend's memories; 5. Carry out the solution — collect photos and make the album; 6. Evaluate the result — if the friend is happy the solution worked, otherwise change the plan.

Question: 3. Take one complex problem from your real life, divide it into small parts and find its solution.

Answer: Example — building a Lego castle: split the castle into parts — base, walls, towers, doors and roof; separate the pieces and colours needed for each part; then build one part at a time and finally join all the parts to complete the castle; dividing makes the big task easy and understandable.

Question: 4. Write a step-by-step algorithm for getting ready for school.

Answer: 1. Start; 2. Wake up; 3. Brush your teeth and bathe; 4. Put on the school uniform; 5. Eat breakfast; 6. Pack the school bag; 7. Go to school; 8.

Stop.

Question: 5. Write a step-by-step algorithm for planting flowers.

Answer: 1. Start; 2. Choose a place for planting; 3. Dig the soil; 4. Put in the seed; 5. Give water; 6. Cover with soil and look after it; 7. Stop.

Question: 6. Design an algorithm that tells whether a given number is even or odd.

Answer: 1. Start; 2. Enter the number; 3. If dividing the number by 2 leaves a remainder of zero, print Even, otherwise print Odd; 4. Stop.

Question: 7. Design an algorithm that finds the sum of three numbers and their average.

Answer: 1. Start; 2. Enter three numbers A, B, C; 3. Find Sum = A + B + C; 4. Find Average = Sum ÷ 3; 5. Print the sum and the average; 6. Stop.

Question: 8. Design an algorithm that finds the product of two numbers.

Answer: 1. Start; 2. Enter two numbers A and B; 3. Find Product = A × B; 4. Print the product; 5. Stop.

Question: 9. Design an algorithm that takes a number and prints it only when that number is 1.

Answer: 1. Start; 2. Enter the number; 3. If the number is 1, print it, otherwise do nothing; 4. Stop.

10. What is meant here by the word "problem"?

(a) A game

✓ (b) A challenge waiting for a solution

(c) A story book

(d) A film

11. Which of the following tasks can be given as an example of a simple problem?

(a) Making a robot

(b) Putting on your shoes

(c) Making a video game

✓ (d) Making a cake

12. What things do we need to make a plan for solving a problem?

(a) Procedures

✓ (b) Facts, input, output

(c) Tools

(d) Detail

13. What is meant by breaking a problem down?

(a) Making it more difficult

✓ (b) Dividing it into small parts

(c) Solving it quickly

(d) Ignoring it

14. What is meant by algorithmic thinking?

✓ (a) Becoming a detective who finds the solution to a secret

(b) Getting lost in thoughts

(c) Watching TV

(d) Playing video games

15. Choose from the given words and fill in the blank: plan, language, part, small, process — Simple problems are ones with _____ requirements.

Answer: small

16. Fill in the blank: Dividing a problem helps us understand each _____ of the problem better.

Answer: part

17. Fill in the blank: An algorithm is a step-by-step _____ for solving a problem.

Answer: plan

18. Fill in the blank: Pseudocode is an easy way of writing an algorithm using everyday _____.

Answer: language.

19. Fill in the blank: In a flowchart, the shapes and arrows guide us during _____ on the instructions or plan.

Answer: the working-through (process) of

Question: 20. Give an example of a simple problem you have faced and tell how you solved it?

Answer: Example — making noodles: pour water, boil it, add the noodles, mix in the seasoning and serve; this problem is simple because it is solved just by joining steps in order and needs no expert.

Question: 21. Why is it necessary to divide a big problem into small parts?

Answer: Because a big problem looks difficult all at once; small parts are easier to understand, solve and test, the facts, input, output and instructions of each part become clear, and then all the parts together give the full solution.

Question: 22. State the benefits of the step-by-step way of solving a problem.

Answer: The step-by-step way is clear and organised; no step is forgotten, each step can be tested separately, a mistake shows up quickly, the work can be repeated, and everyone can easily understand the instructions.

Question: 23. Write about a real-life situation where you can use an algorithm, and make a step-by-step plan for it.

Answer: Situation — making tea: 1. Start; 2. Boil water; 3. Put a tea bag or leaves in the cup; 4. Pour the boiled water into the cup; 5. Leave it for 3 to 5 minutes; 6. Serve with lemon, honey or milk; 7. Stop.

Question: 24. How does a step-by-step plan for solving a problem help reduce the risk of failure?

Answer: Because every step is fixed in advance and clear, no step is forgotten and the order does not get mixed up; each step can be tested, a

mistake can be caught early, and if the result is not good you can go back to step 2 and change the plan; so the chance of failure falls.

Question: 25. Following the problem-solving instructions, write the solution for this situation: Ali has a severe headache and needs to speak to a doctor for treatment.

Answer: 1. Identify the problem — Ali has a severe headache and needs treatment; 2. Gather information — how long the pain has lasted, whether there is fever or other symptoms; 3. Review solutions — rest, water, medicine or going to the doctor; 4. Choose the best solution — consult the doctor; 5. Act — go to the doctor and follow the advice; 6. Evaluate the result — if the pain eases the solution worked, otherwise contact the doctor again.

Question: 26. Using the principle of breaking a problem down, write the small problems for this situation: your mother has asked you to bring household items and you have to go to the shop; the real task is deciding how to buy the household items.

Answer: Small problems: 1. Make a list of which items are needed; 2. Check the money and budget; 3. Decide the route to the shop; 4. Choose the items in the shop and compare prices; 5. Pay and buy the items; 6. Bring the items home and put them away.

Question: 27. Write an algorithm that lets a person enter a number and prints that number's multiplication table up to 10.

Answer: 1. Start; 2. Enter the number N; 3. Set Count = 1; 4. Print N × Count; 5. Set Count = Count + 1; 6. While Count is not more than 10, go back to step 4; 7. Stop.

Question: 28. Write an algorithm that decides whether a student has passed or not; the algorithm takes the marks of four subjects as input and finds the percentage of the marks obtained; if the marks obtained are

more than 60 percent, show the student as passed, otherwise failed.

Answer: 1. Start; 2. Enter the marks of four subjects; 3. Find Total = sum of the four marks; 4. Find Percentage = Total ÷ 400 × 100 (assuming each subject is out of 100); 5. If the percentage is more than 60, print Pass, otherwise print Fail; 6. Stop.

Question: 29. Write an algorithm that adds the first five numbers, that is 1, 2, 3, 4 and 5.

Answer: 1. Start; 2. Set Sum = 0 and Count = 1; 3. Set Sum = Sum + Count; 4. Set Count = Count + 1; 5. While Count is not more than 5, go back to step 3; 6. Print Sum (answer 15); 7. Stop.

Programming / Scratch

1. What does the abbreviation VPL fully stand for?

(a) Graphical programming language

(b) A set of instructions

✓ (c) Visual programming language

(d) A backdrop

2. What is meant by making a program in Scratch?

(a) A list of sprites

(b) A list of backdrops

✓ (c) A combined set of blocks with instructions, used for some task

(d) None of these

3. How many types of code blocks are there in Scratch?

(a) 3

(b) 5

(c) 7

✓ (d) 9

4. Which block is used to start a simple program?

(a) Control block

✓ (b) Event block

(c) Motion block

(d) Looks block

5. Where can code blocks be dragged to give instructions?

(a) Block palette

(b) Backdrop

✓ (c) Script area

(d) Stage

6. Choose from the word box: Wait, If-else, Move block, If block, Repeat — The block used for repetition in Scratch is _____.

Answer: repeats the instructions inside it as many times as the given number.

7. Choose from the word box: Wait, If-else, Move block, If block, Repeat — In Scratch, _____ is the construct used for waiting a short time.

Answer: program for

8. Choose from the word box: Wait, If-else, Move block, If block, Repeat — In Scratch, we check whether a particular condition is true with the _____.

Answer: instructions inside it run.

9. Choose from the word box: Wait, If-else, Move block, If block, Repeat — In Scratch, _____ is the construct used for making a decision.

Answer: condition

10. Choose from the word box: Wait, If-else, Move block, If block, Repeat — In Scratch, we use the _____ to move the sprite forward.

Answer: this block moves

Question: 11. What is programming? Explain it.

Answer: Programming is the process of writing an ordered set of instructions that make a computer do some task; the instructions are in a language the computer can understand and follow to give the wanted result.

Question: 12. State the difference between an algorithm and a program.

Answer: An algorithm is a high-level plan or step-by-step idea for solving a problem, expressed in plain human language or a diagram; a program is the practical form of that plan, written with the rules of a particular programming language, which the computer can translate and run.

Question: 13. Explain sprite, script and stage.

Answer: A sprite is the character or object in a project that is programmed, such as the cat; a script is the set of code-block instructions given to that sprite; the stage is the field or background area where the sprite shows its work.

Question: 14. Explain the nine types of block palette.

Answer: Motion moves the sprite; Looks changes its colour, size and costume; Sound adds sounds; Events starts code on an event; Control handles waiting, repetition and decisions; Sensing gathers information about the surroundings; Operators do maths, comparison and word work; Variables store values; My Blocks lets the user make new blocks of their own.

Question: 15. What is the purpose of the control block?

Answer: The control block manages the flow of the script; with it the program can be paused, a task repeated, and conditional decision instructions given.

Question: 16. In what situations is a conditional statement used?

Answer: When what happens next depends on whether a condition is true or false; if the condition is true one action happens, and if it is false another action happens or nothing happens, as in the body-temperature example or the key-press example.

Question: 17. How is the backdrop changed in Scratch?

Answer: Click the backdrop-choosing option beside the sprite-choosing button, pick a backdrop from the library, bring a random one with Surprise, make your own with the Paint editor, or upload a picture from your computer.

Question: 18. Tell the way to download and install Scratch on your computer.

Answer: Open a browser and go to the download page, choose Direct

download or the store option, run the downloaded file and complete the installation instructions, then double-click the desktop icon to open Scratch; for online work go to the editor address, and an account is needed to save.

Question: 19. Make code that moves the sprite only 10 steps and repeats that action 10 times; which blocks are needed for it?

Answer: The green-flag event block, inside it the control block that repeats 10 times, and inside that the motion block that moves 10 steps; with this the sprite moves 100 steps in total.

Question: 20. Make a round design on the stage in which the sprite moves from left to right and from right to left.

Answer: Under the space-key-pressed event block put four motion blocks that turn 90 degrees each, so the sprite completes a full circle; for back-and-forth movement add the move block and the bounce-on-edge block, and put a wait block between turns so the movement is clearly visible.

Question: 21. How should the teacher guide the students in fixing a bug in the code through the debugging process?

Answer: Follow the six-step cycle: plan the program, write the code, run the program, find the errors, find the cause of the errors, then correct the errors and test the program again.

Question: 22. In the given code the sprite turns 90 degrees four times on the space key, but the turning is not visible; apply the debugging process and fix the bug.

Answer: The bug is that all four turns happen at once with no pause, so they are not visible; put a control Wait 1 seconds block after each turn, then run the program again and check that each turn is seen separately.

Question: 23. How many times will each of the shown loop blocks run?

Answer: The repeat-10 block runs ten times; the repeat-until block keeps running until the given condition is met, such as pressing the space key; the forever block never stops and keeps repeating the instructions inside it.

Digital Citizenship

Question: (Class work 1) Describe how to make a strong password for your email account, using length, letters, a number and a special symbol.

Answer: The password should be at least 8 characters long; start with the first letters of your favourite film's name, then add a favourite digit, then a special symbol, and finally combine the last three letters of your father's name, the first three letters of your mother's name and the first three letters of your own name. Keep a different password for each account, not one for all.

Question: (Class work 2) List the do's and don'ts for keeping yourself safe online as a good digital citizen.

Answer: Do: use a strong password; take permission from parents or teachers; stay kind and polite online; tell an adult about anything wrong. Don't: share personal information; talk to strangers; use a weak password like your name or birth date; bully anyone or post hurtful comments.

Question: (Class work 3) How would you fill in a table to track your daily ICT and social media use?

Answer: Write the time (in minutes) next to each activity: online surfing, watching online films, listening to music, online games, online study, social media use; then add them all up to get total screen time and try to keep it limited.

(MCQ 1) What is the name for copying or sharing copyrighted material or software without permission?

(a) Plagiarism

(b) Copyright

✓ (c) Piracy

(d) Eye strain

(MCQ 2) Digital literacy means the skill of evaluating, accessing and using which kind of material?

✓ (a) Online

(b) On a copy

(c) On paper

(d) On a sketch book

(MCQ 3) What is an illegal act committed using ICT devices and the internet called?

(a) Internet crime

✓ (b) Cybercrime

(c) Internet law

(d) Cyber law

(MCQ 4) What should students do before leaving the laboratory?

(a) Close their accounts

(b) Save their files

(c) Shut down the computers properly

✓ (d) All of the above

(MCQ 5) Where should the top of the computer monitor be?

✓ (a) Slightly below eye level

(b) Slightly above eye level

(c) Perpendicular to eye level

(d) None of the above

(Fill in the blank 1) A good digital citizen has the _____ needed to use information properly.

Answer: knowledge

(Fill in the blank 2) _____ is the law that stops you from using other people's work without their permission.

Answer: Copyright — copyright

(Fill in the blank 3) Your hands should be kept parallel to the floor.

(Fill in the blank 4) A _____ is a person who belongs to a country or nation.

Answer: Citizen — a citizen

(Fill in the blank 5) Spending too much time on screens can harm health, and we may feel _____ and depressed.

Answer: leave us feeling tired

Question: (Detailed question 1) Explain the importance of digital citizenship.

Answer: Digital citizenship teaches us to use the internet and ICT in a safe, responsible and ethical way; it helps us protect our personal information and other people's rights, avoid harmful material and illegal acts, and be good citizens in the online world.

Question: (Detailed question 2) As a digital citizen, how should you behave towards people online?

Answer: Online you should always be kind, polite and respectful; you should not use abusive language or hurtful comments for anyone, bully anyone, or look down on anyone's post with rudeness.

Question: (Detailed question 3) Discuss in detail the ethical issues linked to ICT use while surfing online.

Answer: There are five such issues: (1) misuse of personal information — a name, photo, birth date or home address can fall into wrong hands; (2) spreading false information — unverified news misleads others; (3) hate material — harsh comments and posts based on race, religion or gender; (4) addiction to social media and games — wasting a lot of time and losing focus on studies; (5) online harassment — repeatedly bothering someone,

harming them, or sharing embarrassing material.

Question: (Detailed question 4) Describe the health problems linked to using ICT devices.

Answer: There are six problems: (1) eye strain — staring at a screen too long tires the eyes, so follow the 20-20-20 rule; (2) poor body posture — sitting wrongly causes pain and physical harm; (3) inactive lifestyle and obesity — sitting too long makes the body inactive and increases weight; (4) disturbed sleep — using devices before bedtime disturbs sleep; (5) effects of radiating rays — using devices for too long can cause harm, so screen time should be limited; (6) mental health problems — overuse can cause disappointment, helplessness and depression.

Question: (Detailed question 5) Define the following terms: digital footprint, cybercrime, copyright law, citizenship, cyberbullying.

Answer: Digital footprint: the record of what you do on the internet or social media, including things said about different topics and details shared; Cybercrime: an illegal activity carried out using ICT devices and the internet, such as hacking, fraud, theft of personal information, or online harassment; Copyright law: a law made by the government that protects a creator's new works from being used without permission; Citizenship: a person's belonging to a country or nation; Cyberbullying: repeatedly bothering someone online, harming them, or sharing embarrassing photos or videos.

Question: (Group activity 1) Divide the class into groups; each group prepares a presentation on one health problem caused by overuse of ICT and suggests ways to reduce its risk.

Answer: Each group picks one problem from the chapter, such as eye strain, poor posture, obesity, disturbed sleep, effects of rays, or mental health; then explains its symptoms and ways to prevent it, such as the

20-20-20 rule, sitting correctly, limiting screen time, play and exercise, and switching devices off before bed.

Question: (Group activity 2) How would terms such as plagiarism, piracy and copyright law be explained practically in class?

Answer: The teacher writes these terms on a chart and arranges an activity; students act out the different terms, such as copying someone else's work and calling it their own, downloading material without paying, and taking permission from the owner; this makes the correct use of each term clear.

Business in the Digital Age

Question: Each class group gets the task of setting up a food stall (such as samosas, chaat or salad) in the school ground with a budget of 1000 rupees and writing all expenses in a table — how is this activity completed and what is learned from it?

Answer: Make an expense table (ingredients, utensils, etc.), add up all expenses to get total cost, subtract total cost from total sales income; if income is higher there is a profit, if lower there is a loss — this gives students real experience of running a business.

Question: How do students use digital marketing to publicise the stall from the previous activity?

Answer: Design a poster for the stall (stall name, items, prices and an attractive picture), then post it on the school's Facebook page or WhatsApp group — this develops students' creative skills and shows them that advertising plays a key role in promoting a business.

- What is the main work of a businessperson? (a) Taking part in games (b) Opening and running one's own business (c) Working as an employee for other people (d) Trying to make something new
- What is meant by a business plan? (a) A recipe for making a cake (b) Written guidance for starting a business (c) A list of favourite toys (d) A type of riddle
- What is meant by a new idea or innovation? (a) Doing the same work every day (b) Thinking of new and better ideas (c) Watching TV all day (d) Ignoring problems
- What does a businessperson do when faced with failure? (a) Gives up working (b) Tries again and learns from mistakes (c) Blames other

people (d) Never tries anything new again

What should people who want to start a business do? (a) Wait for someone else to work (b) Work on new and different ideas (c) Watch TV all day (d) Tell others to work for them

Choose from the words below to fill in the blanks: finding an opportunity, profit, technology, businessperson, continuous — Every trade started to earn _____ is called a business.

In the _____ stage, the ability to research and analyse and creative thinking are needed to create something new.

Answer: first stage of

_____ create new markets and expand existing markets, increasing the country's growth.

Answer: Businesspeople — businesspeople

The business process is a _____ process, which helps the business grow.

Answer: never-ending process that

_____ helps businesspeople use tools and gadgets to run their business better.

Answer: Technology — technology

Question: What is meant by a business plan?

Answer: A business plan is a detailed written plan made before starting a business, covering the idea, aim, market research, marketing strategy, operations, funds needed and a summary of the plan; it guides the business to run in an organised way.

Question: How do businesspeople sell their goods or services?

Answer: In the traditional way they sell face-to-face through a shop or stall, or through distributors, and publicise their goods with posters, newspaper, radio and TV adverts; in the digital way they sell through

websites, online shops, social media, email marketing, search engine optimisation and online adverts.

Question: What is meant by identifying an opportunity in the business process?

Answer: Identifying an opportunity is the first stage of the business process, in which the businessperson looks for a new business opportunity or idea and gathers information about it and analyses it; it needs a creative mind and the ability to research and analyse.

Question: In your opinion, what is the greatest quality of a successful businessperson?

Answer: In my opinion the greatest quality is hard work and never giving up, because the chapter explains that a hard-working person overcomes failures and disappointments, seizes opportunities, learns from mistakes and tries again; a good idea alone, without hard work, cannot make any business succeed.

Question: Explain the difference between digital and traditional business.

Answer: Place: traditional business happens in a physical shop, digital business happens online across the whole world; example: traditional — a shop or market, digital — an online shop or app; customer contact: traditional — face-to-face conversation, digital — through online chat and email; publicity: traditional — posters, radio, TV, digital — social media, email, search engine optimisation and online adverts; selling: traditional — directly or through distributors, digital — through e-commerce or subscription; market: traditional — local, digital — global.

Question: What task are students living in different cities given, to tell the world about their city's famous product and help local people sell it? (For example: Hyderabad's bangles, Nawabshah's Sindhi parat)

Answer: Each student chooses their city's famous product, uses their

creativity and social media platforms such as Facebook, Instagram and WhatsApp to introduce that product to the world, and helps local people sell it.

Question: How is the task done of dividing the class into two groups, with one group presenting the success story of a traditional Pakistani businessperson and the other of a digital Pakistani businessperson?

Answer: One group gathers information on the journey of a famous traditional Pakistani businessperson, such as Hakim Muhammad Saeed (Hamdard), and the other on a famous digital Pakistani businessperson, such as Salim Ghauri (NetSol) or Monis Rahman (Rozee), prepares a presentation and presents it to the class.